Advancements in Edge Computing and Autonomous Systems

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1. Executive Abstract

This paper investigates the convergence of distributed edge intelligence and modern autonomous control architectures. Recent advancements in low-power neural processing units (NPUs) enable real-time inference at sub-millisecond latencies, drastically reducing reliance on centralized cloud backbones. We demonstrate an end-to-end framework achieving a 42% decrease in latency and a 35% reduction in wireless bandwidth consumption under adverse network conditions.

2. Problem Formulation and Bandwidth Constraints

Traditional cloud-reliant autonomous pipelines suffer from unpredictable round-trip packet latency and intermittent connectivity drops. In high-speed autonomous vehicular corridors, a 100ms transmission delay can result in catastrophic navigation hazards. Furthermore, transmitting gigabytes of uncompressed point-cloud LiDAR streams saturates local 5G cellular uplinks, increasing operating expenses.

3. Proposed Edge-Native Architecture

Our proposed system introduces a two-tier hierarchical arbitration layer. First, localized telemetry is quantized using 8-bit integer weights and evaluated directly on embedded edge clusters. Second, a speculative compression protocol dynamically filters redundant spatial frames before transmitting metadata to the supervisory coordinator.

4. Experimental Results and Performance Analysis

Extensive benchmarking across 1,000 synthetic test cycles revealed consistent real-time responsiveness. P99 latency dropped from 148ms in pure-cloud baselines to 12.4ms with edge arbitration. Energy efficiency improved by 2.1x per computed trajectory.

5. Conclusions and Future Research

Edge computing represents an essential paradigm shift for mission-critical autonomy. Future research will explore decentralized multi-agent consensus and post-quantum cryptographic verification across ad-hoc wireless mesh links.